Electron track records in radioactive decay. NT2a emulsions were soaked in dilute thorium and uranium nitrate solutions, and electron tracks were recorded at the origin of alpha-tracks. Figs. 1 and 2 show photomicrographs of alpha-tracks of uranium I with beta-tracks probably due to the decay of uranium-X_1. The beta-ray energy (Fig. 1) should, according to the table, correspond roughly to 35 kV. Fig. 3 represents an alpha-track due to thorium-C, and the beta-particle is due to the parent element thorium-B, which has been ejected before the alpha-emission. A second beta-particle is believed to be seen as being ejected from the same origin. This may be attributed to the beta-emission from thorium-C″. Fig. 5 is a photomicrograph-mosaic of a plane-projection of a three-dimensional radiothorium star at the origin of which an electron track is seen. The second electron track visible in this photograph may be the end of a track of an electron of higher energy arising from the same origin.

Auger tracks. While scanning the electron tracks it was observed that some tracks show clumping of grains at the beginning and at the end of tracks (see arrows in Fig. 4). This phenomenon can probably be attributed to the Auger effect. Although the specific ionization is greater at the end of the tracks, in some cases branching of short secondary tracks will be recorded in the emulsion by heavier ionizations at the beginning of the tracks.

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R. H. Herz

Research Laboratories, Kodak, Ltd.,
Wealdstone, Middlesex.

Feb. 1.

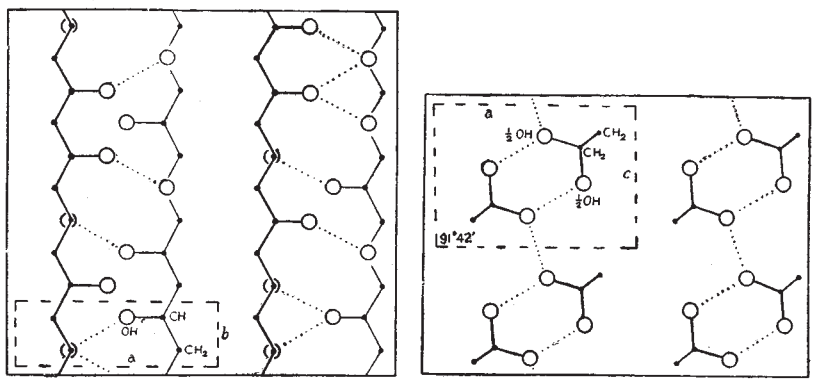
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Crystal Structure of Polyvinyl Alcohol

It has often been said (by me[1] as well as by others) that the molecular repeat distance (from X-ray diffraction photographs) of polyvinyl alcohol—2·52 A.[2]—indicates not only that the chain has a simple plane zigzag configuration, but also that all the hydroxyl groups (in the section of a molecule in one crystallite) lie on the same side of the zigzag plane. This is not necessarily true; so far as X-ray diffraction is concerned, the observed repeat distance is equally compatible with a molecular structure in which hydroxyl groups are randomly placed in left- and right-hand positions. This possibility was not considered until recently; it was assumed that the high degree of crystallinity of polyvinyl alcohol implied regularity of molecular structure. Recently, however, it has been found[3] that interpolymers of ethylene and vinyl alcohol are crystalline; thus hydroxyl groups may replace hydrogen atoms at random on a carbon chain without destroying crystallinity; the high crystallinity of polyvinyl alcohol cannot, therefore, be regarded as evidence against the idea of stereochemical irregularity in this molecule.

To determine whether the hydroxyl groups are randomly or regularly placed, the interpretation of X-ray diffraction patterns of polyvinyl alcohol fibres has been considered in detail. In agreement with Mooney[2], a two-molecule monoclinic unit cell is found to account for the positions of X-ray reflexions; this cell has $a = 7{\cdot}81$ A., $b = 2{\cdot}52$ A., $c = 5{\cdot}51$ A., $\beta = 91°\ 42'$; these figures agree fairly well with those of Mooney, who, however, gives $\beta = 90°$. No structure embodying regular molecules gave correct relative intensities for the X-ray reflexions; all possible arrangements (including that proposed by Mooney) and all possible molecular parameters were considered. But a particular arrangement of molecules with hydroxyl groups randomly placed in left- and right-hand positions accounts in a highly satisfactory way for the entire X-ray diffraction pattern: calculated intensities agree very well with those observed. This arrangement has the symmetry $P2_1/m$ with C (of CH_2) at 0·190 *a*, 0·250 *b*, 0·806 *c*, C (of CH) at 0·277 *a*, 0·750 *b*, 0·708 *c*, half the oxygen atoms at 0·451 *a*, 0·750 *b*, 0·782 *c*, and the other half at 0·267 *a*, 0·750 *b*, 0·450 *c*. The arrangement, which is illustrated in the accompanying projections, may be described as a layer structure; a double layer of molecules is held together by hydroxyl bonds; between one double layer and the next the weaker Van der Waals forces operate. The hydroxyl bond distances (2·8 and 2·9 A.) are greater than in pentaerythritol[4] and resorcinol[5] (2·7 A.) and therefore the bonds are presumably weaker than in those substances; this is perhaps connected with the fact that owing to the random positions of the OH groups, some of the possible bonds are not made at all, while in other cases an OH group is faced with too many possibilities (up to four) of hydroxyl bond formation.

This is an example among high polymers of the sort of randomness which has been reported recently by Clews and Cochran[6] in a monomeric crystal (2-amino-4-methyl-6-chloro-pyrimidine); but in view of the difference of size of the hydroxyl group and the hydrogen atom, polyvinyl alcohol is a more extreme case. The implications in the theory of crystallization (or non-crystallization, as the case may be) of high polymers have been referred to in an earlier communication[3]. Here, I need only remark that polyvinyl acetate, from which polyvinyl alcohol is prepared, probably has a corresponding (stereo-



c-Projection *b*-Projection

Dotted lines, hydrogen bonds

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chemically random) molecular structure, but, unlike polyvinyl alcohol, does not crystallize; this is presumably because the difference of size between acetate groups and hydrogen atoms is much too great for them to occupy equivalent positions in a crystal structure.

C. W. Bunn

Research Department,
Imperial Chemical Industries, Ltd.,
Plastics Division,
Welwyn Garden City, Herts.
Feb. 26.

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The Tollens Naphthoresorcinol Reaction for Uronic Acids

The mechanism of colour production in the Tollens test[1] for glucuronic acid is not definitely known, and since this reaction is frequently used for the determination of glucuronic acid in biological materials[2], an attempt should be made to understand its chemistry. Some of our preliminary results to this end are now reported here.

Since the reaction is carried out by heating naphthoresorcinol (1 : 3-dihydroxynaphthalene) and glucuronic acid in 5–6 *N* hydrochloric acid, furfural may be formed which may condense with naphthoresorcinol to give a pigment. We find, however, that furfural gives a different and less intense colour than glucuronic acid.

Mayer, Bloch and Chaffee[3] in a brief abstract suggested that the pigment of the Tollens test may be a xanthene derivative. Our observations (see below) appear to support this view, and the following mechanism is suggested for the Tollens reaction (cf. the formation of pyronine and phthaleine dyes).

Some of our observations were :

(*a*) Heating naphthoresorcinol, m.p. 119°, and *d*-glucurone, m.p. 176° (or *d*-galacturonic acid), together in the dry state at a low temperature, 100–120°, yields the same dark-blue solid as obtained in the Tollens reaction. In ethanol the blue solid obtained either way shows spectral absorption bands at 595, 335, 302 and 243 mμ.

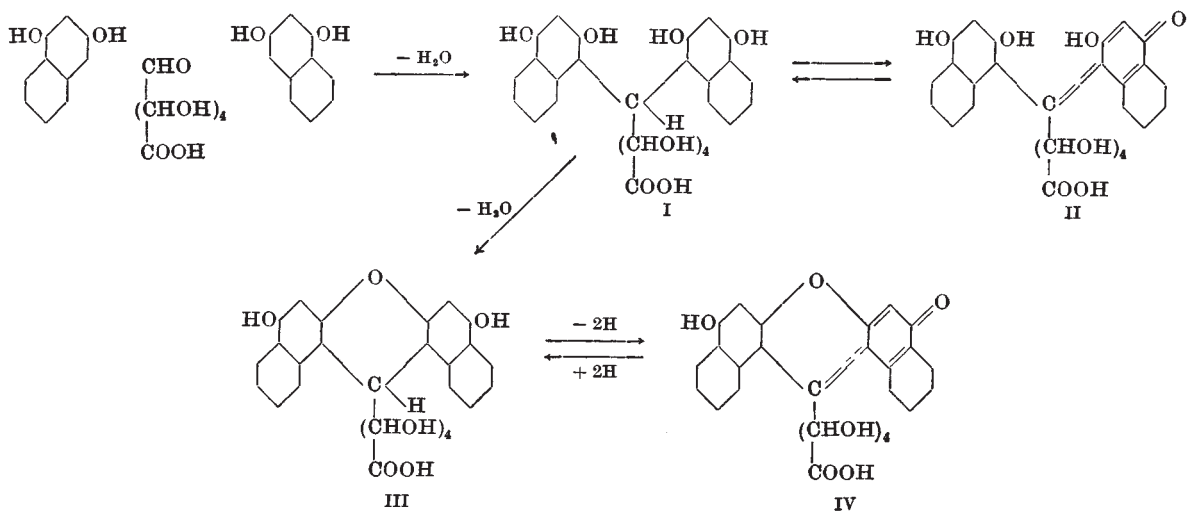
(*b*) If dry naphthoresorcinol and *l*-menthyl-glucuronide are heated together, no blue solid is obtained unless crystalline oxalic acid is added to the melt. This experiment suggests that the aldehydic group of the uronic acid must be free for pigment formation.

(*c*) The solid pigment is soluble in dilute alkali, from which it is precipitated by acid as dark-blue flocks, that is, it is acidic.

(*d*) A solution of the pigment is readily decolorized by sodium hydrosulphite, forming the leuco dye I or III, and the colour can be restored (to the quinones II or IV) by an oxidizing agent such as chloramine-*T*.

(*e*) If naphthoresorcinol is replaced in the Tollens reaction by monocyclic *m*-dihydric phenols such as resorcinol or orcinol, only very weak colours are obtained unless an oxidizing agent is also present. Thus if orcinol and glucurone are heated in 5–6 *N* hydrochloric acid, on extraction with amyl alcohol a weak green colour is obtained; but if the reaction is carried out in the presence of chloramine-*T*, a very intense green results. To obtain an intense colour with naphthoresorcinol an external oxidizing agent does not appear to be necessary, although Mayer, Bloch and Chaffee[3] state that the colour intensity is increased by potassium persulphate. It should be noted that the two possible formulæ, II and IV, suggested for the naphthoresorcinol pigment could form several resonance structures; these would tend to stabilize the pigment and render its production from the leuco-form relatively easy. This suggests that *m*-diols of polycyclic hydrocarbons such as anthracene-1 : 3-diol may be even better than naphthoresorcinol for detecting uronic acids, for in such compounds more resonance structures are possible.

We believe that the pigment of the Tollens reaction could be either a dinaphthylmethane (II) or a xanthene derivative (IV). We have found the isolated pigment difficult to purify, and we cannot as yet rely on elementary analyses for confirmation of the structures suggested. Analysis of a partially purified



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